

lab 12(Trigger)

Task 01:

Create Employee Table and insert data:

```
#Create Employee Table
CREATE TABLE Employee (
    Id INT PRIMARY KEY,
    Name VARCHAR(45),
    Salary INT,
    Gender VARCHAR(12),
    DepartmentId INT
);
#insert
INSERT INTO Employee (Id, Name, Salary, Gender, DepartmentId) VALUES
(1, 'Steffan', 82000, 'Male', 3),
(2, 'Amelie', 52000, 'Female', 2),
(3, 'Antonio', 25000, 'Male', 1);
```

Create Audit Table

```
CREATE TABLE Employee_Audit (
    Id INT auto_increment PRIMARY KEY,
    Audit_Action VARCHAR(255)
);
```

Create Insert Trigger

```

DELIMITER //
• CREATE TRIGGER trInsertEmployee
  AFTER INSERT ON Employee
  FOR EACH ROW
  BEGIN
    INSERT INTO Employee_Audit (Audit_Action)
    VALUES (
      CONCAT(
        'New employee with Id = ', NEW.Id,
        ' added at ', NOW()
      )
    );
  END //
DELIMITER ;

```

Test Insert Trigger

```

#Test Insert Trigger
INSERT INTO Employee (Id, Name, Salary, Gender, DepartmentId)
VALUES (5, 'Marco', 47000, 'Male', 2);
SELECT * FROM Employee_Audit;

```

Delete Trigger

```

DELIMITER //
• CREATE TRIGGER trDeleteEmployee
  AFTER DELETE ON Employee
  FOR EACH ROW
  BEGIN
    INSERT INTO Employee_Audit (Audit_Action)
    VALUES (
      CONCAT(
        'Employee with Id = ', OLD.Id,
        ' deleted at ', NOW()
      )
    );
  END //
DELIMITER ;

```

Test Delete Trigger:

```

DELETE FROM Employee WHERE Id = 1;
SELECT * FROM Employee_Audit;

```

Task02:

Create the Main Table

```
CREATE TABLE Book01 (  
    BookID INT PRIMARY KEY auto_increment,  
    Title NVARCHAR(100),  
    Author NVARCHAR(100),  
    Price DECIMAL(10,2)  
);
```

Create the Audit Table

```
CREATE TABLE Books_Audit (  
    AuditID INT PRIMARY KEY auto_increment,  
    Audit_Action NVARCHAR(200),  
    Audit_Time DATETIME  
);
```

After INSERT Trigger

```
DELIMITER //  
CREATE TRIGGER trBooks_Audit_Insert  
AFTER INSERT ON Books  
FOR EACH ROW  
BEGIN  
    INSERT INTO Books_Audit (Audit_Action)  
    VALUES (CONCAT('Inserted Book with ID = ', NEW.BookID, ' at ', NOW()));  
END //  
DELIMITER ;
```

After UPDATE Trigger

```
DELIMITER //  
CREATE TRIGGER trBooks_Audit_Update  
AFTER UPDATE ON Books  
FOR EACH ROW  
BEGIN  
    INSERT INTO Books_Audit (Audit_Action)  
    VALUES (CONCAT('Updated Book with ID = ', NEW.BookID, ' at ', NOW()));  
END //  
DELIMITER ;
```

After DELETE Trigger

```
DELIMITER //
CREATE TRIGGER trBooks_Audit_Delete
AFTER DELETE ON Books
FOR EACH ROW
) BEGIN
    INSERT INTO Books_Audit (Audit_Action)
    VALUES (CONCAT('Deleted Book with ID = ', OLD.BookID, ' at ', NOW()));
- END //
DELIMITER ;
```

Task 03:

This table stores bank account details

```
CREATE TABLE Account01 (
    AccountID INT PRIMARY KEY AUTO_INCREMENT,
    AccountHolder VARCHAR(100) CHARACTER SET utf8mb4,
    Balance DECIMAL(10,2),
    AccountType VARCHAR(20) CHARACTER SET utf8mb4
);
#Insert Sample Data
INSERT INTO Account01 (AccountHolder, Balance, AccountType)
VALUES ('Ali Khan', 5000.00, 'Savings'),
('Sara Ahmed', 10000.00, 'Current'),
('John Doe', 7000.00, 'Saving');
```

Transaction_Audit Table

```
CREATE TABLE Transaction_Audit1 (
    AuditID INT PRIMARY KEY AUTO_INCREMENT,
    Audit_Action VARCHAR(255) CHARACTER SET utf8mb4,
    AccountID INT,
    ActionDate DATETIME
);
```

Login_Audit Table

```
CREATE TABLE Login_Audit (  
    LogID INT PRIMARY KEY AUTO_INCREMENT,  
    LoginName VARCHAR(100) CHARACTER SET utf8mb4,  
    LoginTime DATETIME  
);  
-- MySQL Workbench (AFTER TRIGGER)
```

DML Trigger

After INSERT Trigger

```
DELIMITER //  
CREATE TRIGGER trAccount_Audit_Insert  
AFTER INSERT ON Accounts  
FOR EACH ROW  
BEGIN  
    INSERT INTO Transaction_Audit (Audit_Action, AccountID, ActionDate)  
    VALUES (  
        CONCAT('New Account Created with ID = ', NEW.AccountID),  
        NEW.AccountID,  
        NOW()  
    );  
END //  
DELIMITER ;
```

After UPDATE Trigger

```
DELIMITER //  
CREATE TRIGGER trAccount_Audit_Update  
AFTER UPDATE ON Accounts  
FOR EACH ROW  
BEGIN  
    INSERT INTO Transaction_Audit (Audit_Action, AccountID, ActionDate)  
    VALUES (  
        CONCAT('Account Details Updated for ID = ', NEW.AccountID),  
        NEW.AccountID,  
        NOW()  
    );  
END //  
DELIMITER ;
```

After DELETE Trigger

```
DELIMITER //
CREATE TRIGGER trAccount_Audit_Delete
AFTER DELETE ON Accounts
FOR EACH ROW
BEGIN
    INSERT INTO Transaction_Audit (Audit_Action, AccountID, ActionDate)
    VALUES (
        CONCAT('Account Deleted with ID = ', OLD.AccountID),
        OLD.AccountID,
        NOW()
    );
END //
DELIMITER ;
```